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Hamza Mohammed Hassanain

Software Engineer

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EDUCATION

B.Sc. Computer and Information Sciences, *Suez Canal University*
AWS Certified Solutions Architect – Associate, *Amazon Web Services*

2022 – 2026
credly.com/verify

EXPERIENCE

Founding Software Engineer Nov 2025 – Present
Repovive (YC F’26) Full Time – United States, Remote
Platform for elite problem solvers (10,000+ users): contests, distributed judging, AI problem authoring, and AI voice interviews.

- **Founding engineer**, longest-tenured on the team: built core systems from early stage to YC and 10,000+ users.
- Led design of the distributed judge (judge1 on **Judge0**); scaled submissions **10x (50 to 500+ concurrent)**.
- Cut infrastructure cost **95% (\$600 to \$25/mo)** by migrating resource-heavy tasks from serverless to Docker/Nginx/Redis.
- Isolated **20+ languages** in resource-limited Docker images; redesigned the DB layer for **50MB+** contest test cases.
- Built AI voice interviews (**LiveKit**): recruiter, behavioral, and coding agents run from versioned, admin-authored definitions.
- Shipped partner screenings: single-use invite links, atomic grading claims, **MongoDB transactions**, per-IP rate limits.
- Hardened agents: answer keys served via authenticated agent routes, candidate text fenced against **prompt injection**.
- Moved CI to **Vercel** when Actions quota ran out: one fail-fast pipeline (format, typecheck, test, build), no prod secrets.
- Root-caused live-contest judge outages (crash-looping mid-contest deploys); drove zero-downtime deploys, shipped 10 fixes.
- Built a real-time collaborative dev space using **WebSockets** and **Yjs CRDTs** over a Monaco live-syncing editor.
- Co-founded **Build v2**, an AI problem-authoring IDE: 28-tool agent pipeline, execution locks, 10-minute streaming sessions.
- Tech: TypeScript, Next.js, Python, LiveKit, AWS, Docker, Nginx, Redis, MongoDB, Vercel, Yjs.

OPEN SOURCE CONTRIBUTIONS

- **GitLab Go SDK (client-go)**: typed group integrations – Slack, Jira, MS Teams, Mattermost, Harbor, +3 | **!2692, !2691, !2675, !2670**.
- **GitLab Terraform provider**: new Harbor & Mattermost integration resources + group SAML attribute | **!2881, !2909, !2852**.
- **GitLab monorepo**: `Json.safe_parse` hardening and Vue lint compliance | **!221163, !221158**.
- **LLVM / Clang**: `constexpr X86 cvtpd2ps, AArch64/ARM CLS` → `ISD`: `:CTLS`, `LLDB SWIG tests` | **#169980, #178885, #178430, #173473**.
- **Apple Swift NIO**: idempotent `NIOFileSystem` directory creation | **#3410**.

PROJECTS

Autonomous AI Tutor & Curriculum Engine, Graduation Project github.com/sigma-loop
• Led backend & DevOps for an 8-person team building an on-demand AI course, lesson, and challenge generator.
• Graded code in a Judge0 sandbox and math proofs via a confidence-scored LLM; owned all CI/CD and AWS deployments.

Competitive-Programming Judge — AWS Reference Architecture github.com/HamzaHassanain/CP-Judge-AWS
• Designed an AWS reference architecture: Judge0 `isolate` on ECS-on-EC2, Fargate orchestrator, Spot ASG on queue depth.
• Automated post-round cheating detection with Step Functions, AWS Batch, Lambda/Bedrock, and Athena.

Polyman CLI npmjs.com/package/polyman-cli | github.com/HamzaHassanain/polyman
• Built a TypeScript CLI automating the Codeforces Polygon workflow: test generation, validation, and direct API sync.

Systems in C++ — HALlocator & cppress HALlocator | cppress
• Built a Red-Black-Tree best-fit memory allocator (**O(log n)**) and an epoll-driven C++ web framework, both from scratch.

TECHNICAL WRITING

- Authored ["What Every Programmer Should Know About Memory"](#), a series on CPU caches, virtual memory, and optimization.
- Published ["I Built a Backend Web Framework from Scratch in C++"](#), covering non-blocking I/O, HTTP parsing, and thread pools.

ACHIEVEMENTS

ICPC ECPC & ACPC — Competitive Programming icpc.global/YC7Z02WQCPIB
17th of 500 teams at ECPC (2024); two-time ACPC regional qualifier (2024, 2025), top 250 in the Africa & Arab region.

SKILLS

Cloud & DevOps	AWS, Docker, Nginx, Redis, Terraform, CI/CD (GitHub Actions), DigitalOcean, Linux
Backend & Full-Stack	TypeScript, Node.js, Express.js, Next.js, React, MongoDB, PostgreSQL, REST APIs, WebSockets, Yjs (CRDTs)
Languages & Systems	C++, Go, Rust, Python, SQL; System Design, Distributed Systems, Message Queues